MILITARY SPECIFICATION

LUBRICATION CHARTS, PLATES, AND MANUALS; PREPARATION AND PRODUCTION OF

1. SCOPE

1.1 Scope. - This specification covers the detail requirements for the preparation and production of lubrication instructions media.

1.2 Classification. - Lubrication instruction media shall be of the following types, as specified (see 6.2):

Type I - Paper, plastic enclosed
Type II - Plate, metal, photographic
Type III - Decal
Type IV - Publication element
Type V - Pamphlet

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and other publications, of the issues in effect on the date of the invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-78	- Plastic-Material, Laminated, Thermosetting
MIL-D-8634	- Decal, Elastomeric Pigmented Film, for Use on Exterior Surfaces
MIL-D-8635	- Decals, for Use on Interior Surfaces
MIL-P-15024	- Plates, Identification-Information and Marking for Identification of Electrical, Electronics, and Mechanical Equipment

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MIL-P-21630 - Production and Preparation of Illustrations for Marine Corps Publications

MIL-T-22689 - Technical Manuals, Production of

STANDARDS

Military

MIL-STD-12 - Abbreviations for Use on Drawings and in Technical-Type Publications

MIL-STD-15 - Electrical and Electronic Symbols

MIL-STD-17 - Mechanical Symbols

MIL-STD-33 - Lubrication Instructions, Preparation and Presentation of

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-290 - Packaging, Packing, and Marking of Petroleum and Related Products

PUBLICATIONS

U.S. Marine Corps

SG-1 - Technical Manuals, Style Guide for the Preparation of

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

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Department of Defense

Federal Supply Catalog C4-1 - Petroleum, Petroleum Base Products, and Related Materials

Industrial Security Manual for Safeguarding Classified Information (Attachment to DD Form 441)

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

Official Classification Committee

Uniform Freight Classification Rules

(Applications for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

American Trucking Associations

National Motor Freight Classification Rules

(Applications for copies should be addressed to the American Trucking Associations, 1616 "P" Street, N.W., Washington, D.C.)

3. REQUIREMENTS

3.1 Design, general. - The general design for the lubrication instruction media produced in accordance with this specification shall meet all the requirements specified herein. The design for each Type I, II, or III medium shall be executed in the form of a layout suitable for use in its construction (see 3.4). The design of a Type IV medium shall be executed in the form of a manuscript marked for printing, and illustrations marked for reproduction. The design for a Type V medium shall be executed in the form of a marked dummy construction guide, with manuscript marked for printing, and illustrations marked for reproduction.

3.1.1 Type I media. - Every Type I medium (paper, plastic enclosed) shall be designed as a graphic chart containing illustrations and instructional data. It shall conform with the design illustrated in figures 1 and 2. The principal illustration shall be the focal point

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of the design which shall conform in its detail to the provisions of Specification MIL-P-21630. Whenever practical, the ratio of width to length should be maintained at 1 to 1.25. The overall dimensions of a Type I medium shall not exceed 15 inches by 25 inches.

3.1.2 Type II media. - Unless otherwise specified, every Type II medium (plate, metal, photographic), shall be designed to the same requirements as a Type I medium, except that the overall dimensions shall not exceed 12 inches by 20 inches.

3.1.2.1 Individual application media. - Whenever it is determined that a Type II medium is required for an application, other than that for which the media covered by 3.1.2 are suitable, it shall be considered as an individual application. In such cases the design of the medium shall be specifically related to its application, and in accordance with the provisions of Specification MIL-P-21630.

3.1.3 Type III media. - Unless otherwise specified, every Type III medium (Decal) (see 6.2), shall be designed to the same requirements as a Type I medium, except that the overall dimensions shall not exceed 10 inches by 15 inches.

3.1.3.1 Individual application media. - Whenever it is determined that a Type III medium is required for an application, other than that for which the media covered by 3.1.2 are suitable, it shall be considered as an individual application. In such cases, the design of the medium shall be specifically related to its application, and in accordance with the provisions of Specification MIL-P-21630.

3.1.3.2 Mounting instructions. - Type III media which require specific handling in mounting or in location shall be designed to permit printing of such data on the paper backing sheet.

3.1.4 Type IV media. - Every Type IV medium shall be designed for publication as a chapter in a technical manual, in accordance with the provisions of the Marine Corps Style Guide for the Preparation of Technical Manuals, SG-1.

3.1.5 Type V media. - Every Type V medium shall be designed as a complete publication. Its elements shall conform to the applicable designs illustrated in figures 3, 4, 5, and 6, herein, and of figures 2 and 11 of Specification MIL-T-22689.

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3:2 Contents. - The content of lubrication instruction media prepared in accordance with this specification shall meet the requirements specified herein.

3.2.1 Type I media. - A Type I medium, commonly termed a "laminated chart," is illustrated in figures 1 and 2. The encircled numerals in the figures correspond to the "item" numbers cited in paragraphs 3.2.1.1 through 3.2.1.17.

3.2.1.1 Heading (Item No. 1). - A Type I medium shall have a heading consisting of the words "LUBRICATION INSTRUCTION" (see 3.4.5.3).

3.2.1.2 Numerical designator (Item No. 2). - A Type I medium shall have a numerical designator. This designator includes the publication number, effective date, and when applicable, the supersession notice. Data for the numerical designator will be supplied by the procuring activity. (See 6.2.)

3.2.1.3 Title (Item No. 3). - A Type I medium shall show as a title the standard nomenclature of the equipment to which it applies. This data will be furnished by the procuring activity (see 6.2).

3.2.1.4 References (Item No. 4). - Pertinent references shall be shown in Type I media. When multiple references are cited they shall be identified in series as follows: (a), (b), (c), etc.

3.2.1.5 General notices (Item No. 5). - Information of a strictly general nature, applying to the equipment as a whole shall be shown as general notes.

3.2.1.6 Illustrations (Item No. 6). - Each Type I medium shall contain at least one illustration of the item. Additional illustrations may be employed, when necessary to properly locate and identify specific points or areas. Such additional illustrations shall be identified by subtitles positioned immediately below them.

3.3.1.7 Lubrication point data (Item No. 7). - Every lubrication point in a Type I medium shall be associated with data showing the point name, the lubricant symbol, and the lubrication interval. The data for each point shall be arranged in a horizontal line positioned on the left or right side of the illustration as appropriate. The point names, symbols, and intervals shall be so spaced that when the chart is viewed as a whole, they appear respectively in vertical columns.

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3.2.1.8 Lubrication point indicators (Item No. 8). - Each lubrication point shall be located on the illustration by connecting it to its point data line by means of an arrow-headed-leader line as shown in figures 1 and 2.

3.2.1.9 Correspondent point indicators (Item No. 9). - When a lubrication point visible on the illustration is matched by an identical counter-point on the opposite (hidden) view of the equipment, the leader line used shall be a dotted line in lieu of a solid line.

3.2.1.10 Point notes (Item No. 10). - Special information, which may be presented in brief form shall be arranged as a point note, so positioned on the chart as to appear directly under the data line of the point to which it applies. Whenever the special information may not be so presented, the point note shall show by reference where it may be found.

3.2.1.11 Lubricant symbols (Item No. 11). - Lubricant symbols used in Type I media shall be selected from those shown in Federal Supply Catalog C4-1 or in Standard MIL-STD-290. In the event a symbol has not been assigned to a required lubricant, the Contractor shall obtain necessary direction from the procuring activity.

3.2.1.12 Lubrication intervals (Item No. 12). - Lubrication intervals used in Type I media shall conform to those shown in Standard MIL-STD-33. Where a required interval is not shown in MIL-STD-33, or where it is irregular, or variable, such interval is shown only in the form of a note in the Type I medium (see 3.2.1.14) with an appropriate reference as a point note (see 3.2.1.10).

3.2.1.13 Key (Item No. 13). - Every Type I medium shall include a key. This key shall be arranged in the form of a chart containing closed columns and subcolumns, as required. A typical arrangement is illustrated in figure 1. The key shall contain the following basic information in column arrangement beginning at the left edge:

a. An index column listing by symbol all the lubricants used with the equipment item.

b. A name-use column showing the name of each lubricant indexed and its application in the equipment item.

c. A capacity column, when the number of lubricant reservoirs, such as gear boxes, is large.

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d. An "expected temperatures" column, divided into three subcolumns covering ranges as shown in the figure.

e. When applicable, a column with the wording "For Arctic operation, see equipment technical manual." This column is to read from the right-hand edge of the key.

f. An interval column, listing all the interval symbols used in the instruction, and the time periods they indicate.

g. A note column, when required.

(Column, or line spanning when practical may be arranged as indicated in figure 1.)

3.2.1.14 Notes (Item No. 14). - Notes shall be used, as necessary, to expand lubrication point data (see 3.2.1.7), or to explain irregular or variable intervals (see 3.2.1.12). They may also be used to present specific information not properly a part of the general notices (see 3.2.1.5).

3.2.1.15 Authentication (Item No. 15). - Every Type I medium shall bear an authentication notice. This notice shall be positioned as illustrated in figure 2. When the Type I medium is composed of more than one page, the authentication notice shall appear on the last page only. The wording required for the notice will be furnished by the procuring activity.

3.2.1.16 Negative number (Item No. 16). - Arrangement shall be made for the Marine Corps negative number to be applied in accordance with Specification MIL-P-21630.

3.2.1.17 Sheet numbering (Item No. 17). - A Type I medium which is completely contained in one page shall not bear a sheet number. When more than one sheet is required, each sheet shall be numbered to indicate its position in the total number of sheets in the series, e.g., "Sheet 1 of 6," "Sheet 2 of 6," etc.

3.2.2 Type II media. - Unless otherwise specified (see 6.2) the content of Type II media prepared in accordance with this specification shall be identical with that of Type I media (see 3.2.1).

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3.2.2.1 Individual application media. - Whenever an individual application Type II medium is required (see 3.1.2.1), its content shall be based upon the specific application, prepared in accordance with Specification MIL-P-21630.

3.2.3 Type III media. - Unless otherwise specified (see 6.2) the content of Type III media prepared in accordance with this specification shall be identical with that of Type I media (see 3.2.1).

3.2.3.1 Individual application media. - Whenever an individual application Type III medium is required (see 3.1.3.1), its content shall be based upon the specific application, prepared in accordance with provisions of Specification MIL-P-21630.

3.2.3.2 Mounting instructions. - When a Type III medium requires specific handling for locating or mounting, suitable data shall be incorporated as part of the content of the medium. It shall be prepared for printing on the paper backing sheet.

3.2.4 Type IV media. - Type IV media shall be prepared in accordance with the provisions of the Marine Corps Style Guide for the Preparation of Technical Manuals, SG-1. A Type IV medium shall be prepared as a complete chapter and shall include the following, arranged in sections and paragraphs, as appropriate:

a. Scope of the chapter

b. General lubrication information

c. Care, handling, and storage of lubricants

d. Lubrication points and levels

e. Service intervals

f. Abnormal operating-condition instructions

g. Chart (when specified) identical in arrangement to that required for a Type I medium (see 3.2.1), except authentication (3.2.1.15) and sheet numbering (3.2.1.17), not required. Charts used in a Type IV medium shall be assigned appropriate figure numbers.

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3.2.5 Type V media. - A Type V medium shall be prepared as a complete publication, in general accordance with the provisions of the Marine Corps Style Guide for the Preparation of Technical Manuals, SG-1. Emphasis shall be placed upon the use of illustrative material, prepared in accordance with Specification MIL-P-21630, and arranged as shown in figures 4, 5, and 6 hereof. Additional information, as required to expand or enlarge that shown in the illustrations, and information of a general nature shall be arranged as text matter. The elements of a Type V medium include those listed in the subparagraphs following.

3.2.5.1 Cover, front. - Shall be arranged as shown in figure 3. Data concerning the numerical designator, title, and effective date shall be as furnished by the procuring activity (see 6.2).

3.2.5.2 Letter of promulgation. - Shall be as furnished by the procuring activity.

3.2.5.3 List of effective pages. - Shall be in accordance with the format illustrated in figure 4 of Specification MIL-T- 22689. Required only for a pamphlet of 10 or more pages.

3.2.5.4 Record of changes. - Shall be in accordance with the format illustrated in figure 5 of Specification MIL-T-22689. Required only for a pamphlet of 10 or more pages.

3.2.5.5 Safety summary. - When required, shall be in accordance with the format illustrated in figure 6 of Specification MIL-T-22689.

3.2.5.6 Table of contents. - Shall be in accordance with the format illustrated in figure 7 of Specification MIL-T-22689. Required only for a pamphlet of 10 or more pages.

3.2.5.7 List of illustrations. - Shall be in accordance with the format illustrated in figure 8 of Specification MIL-T-22689. Required only when 10 or more illustrations are included.

3.2.5.8 List of tables. - Shall be in accordance with the format illustrated in figure 10 of Specification MIL-T-22689. Required only when 10 or more tables are included.

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3.2.5.9 Text or body. - Shall be prepared in accordance with the format illustrated in figure 11 of Specification MIL-T-22689, and the requirements of Style Guide, SG-1, except that no chapter or section divisions shall be used. The standard text title shall be "Lubrication Instructions"; the text shall be arranged in paragraphs and subparagraphs. When paragraph or subparagraph headings are required, they shall be arranged to be run in line.

3.2.5.10 Illustrations. - Shall be prepared in accordance with the provisions of Specification MIL-P-21630, and in agreement with the format illustrated in figures 4, 5, and 6.

3.2.5.11 Cover, back. - Shall be prepared in accordance with the format illustrated in figure 2 of Specification MIL-T-22689.

3.2.6 Abbreviations and symbols, use of. - The use of abbreviations and symbols will, in general, be governed by the requirements of the Style Guide for the Preparation of Technical Manuals, SG-1. When limitations of space in Types I, II, and III make the use of abbreviations and symbols necessary, those used shall conform to standards MIL-STD-12, MIL-STD-15, MIL-STD-17, and MIL-STD-290.

3.3 Material. - Material used in the production of lubrication instruction media shall be as specified herein. Material not definitely specified shall be of a good commercial quality, entirely suitable for the purpose. Material shall be free from all defects and imperfections that might affect the suitability of the finished product.

3.3.1 Paper stock

3.3.1.1 Covers. - The paper stock used for front and back covers of Type V media shall conform to the requirements of Specification MIL-T-22689.

3.3.1.2 Charts, text, and body. - The paper stock for charts produced for Type I media and for text and body of Type IV and Type V media shall conform to the requirements for body and text paper set forth in Specification MIL-T-22689.

3.3.1.3 Fold-out or foldup leaves. - The paper stock used for fold-out or foldup leaves used in Type IV and Type V media shall conform to the requirements of Specification MIL-T-22689.

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3.3.2 Plastic. - Plastic used for Type I media shall conform to the requirements of Specification MIL-P-78A, Type GCP-H, and shall have a thickness of 8 mils.

3.3.3 Plates. - Plates used for Type II media shall conform to the requirements of Specification MIL-P-15024, Type H, Aluminum, and shall have a thickness of 0.020 inch.

3.3.4 Decals. - The decals used for Type III media shall conform to the requirements of Specification MIL-D-8634 or MIL-D-8635, as specified (see 6.2). For mounting, decals shall not require the use of cements, tools, or materials not readily available to using personnel.

3.3.5 Ink. - Unless otherwise specified (see 6.2) all ink used for Types I, III, IV and V shall be of one single black color only.

3.4 Construction. - The construction of lubrication instruction media produced in accordance with this specification, shall meet all the requirements specified herein.

3.4.1 Type I media. - A Type I medium shall consist of a lubrication instruction composed and made up in accordance with a layout encompassing a design as required in 3.1.1 and a content as required in 3.2.1. It shall be printed in black ink (3.3.5) on white paper (3.3.1.2) and totally enclosed by and heat sealed within a plastic envelope (3.3.2). The plastic envelope shall extend not less than 1/8 inch beyond the edges of the paper sheet on all sides. All corners of the finished envelope shall be rounded on a radius appropriate to the overall size of the medium.

3.4.2 Type II media. - A Type II medium shall consist of a lubrication instruction composed and made up in accordance with a layout encompassing a design as required in 3.1.2, and a content as required in 3.2.2. It shall be permanently imprinted on an aluminum plate through a photographic process (see 3.3.3). Unless otherwise specified (see 6.2) the image shall have black lines on a light background. When mounting by screws or rivets is specified (see 6.2), each medium shall have punched or drilled mounting holes at the locations and in the sizes indicated on the layout. A sufficient quantity of suitable aluminum mounting screws or rivets shall be furnished as a component of each Type II medium. When mounting by adhesive is specified (see 6.2), a sufficient quantity of adhesive,

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suitable for mounting the Type II medium to the equipment item shall be furnished as a component of the medium. The adhesive shall be noncorrosive, nontoxic, and waterproof when dry. Its use shall not require special or unusual tools, equipment, or techniques.

3.4.3 Type III media. - A Type III medium shall consist of a lubrication instruction composed and made up in accordance with a layout encompassing a design as required in 3.1.3, and a content as required in 3.2.3. It shall be imprinted on a decal of the interior or exterior type (3.3.4) as specified (see 6.2) in black ink on a background of the color as specified (see 6.2). Specific locating or mounting instructions shall be printed on the paper, backing sheet. Such instructions shall be printed in 10 point modern type in an appropriate line length not exceeding 37 1/2 picas.

3.4.4 Type IV media. - A Type IV medium shall consist of a lubrication instruction composed and made up in accordance with a design as required in 3.1.4, and a content as required in 3.2.4. It shall be produced as a part of the publication for which it was prepared and shall be integrated therewith.

3.4.5 Type V media. - A Type V medium shall consist of a lubrication instruction composed and made up in accordance with a dummy construction guide encompassing a design as required in 3.1.5 and a content as required in 3.2.5. It shall be produced as a complete publication, in accordance with the requirements specified herein.

3.4.5.1 Method of reproduction. - All Type V media produced in accordance with this specification shall be printed by the Photolithographic offset process.

3.4.5.2 Materials. - Materials employed in the printing and binding of Type V media in accordance with this specification shall meet all the requirements of paragraph 3.3.

3.4.5.3 Type sizes and faces. - In accordance with paragraph 3.3.1 of Specification MIL-T-22689.

3.4.5.4 Type page size. - In accordance with paragraph 3.3.2 of Specification MIL-T-22689.

3.4.5.5 Column measures. - In accordance with paragraph 3.3.3 of Specification MIL-T-22689.

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3.4.5.6 Paragraph indentions. - In accordance with paragraph 3.3.4 of Specification MIL-T-22689.

3.4.5.7 Notes and cautions. - In accordance with paragraph 3.3.5 of Specification MIL-T-22689.

3.4.5.8 Warnings. - In accordance with paragraph 3.3.6 of Specification MIL-T-22689.

3.4.5.9 Security indicia. - In accordance with paragraph 3.3.7 of Specification MIL-T-22689.

3.4.5.10 Type composition. - In accordance with the applicable portions of paragraph 3.4 of Specification MIL-T-22689, modified in accordance with the applicable illustrations of this specification.

3.4.5.11 Arrangement or makeup. - The arrangement or make-up of a Type V medium shall be in accordance with the applicable portions of paragraph 3.5 of Specification MIL-T-22689 as modified by the arrangement shown in the dummy construction guide (3.4.5).

3.4.5.12 Binding. - The binding of Type V media shall be in accordance with the applicable portions of paragraph 3.6 of Specification MIL-T-22689.

3.4.6 Reproducibles media. - Unless otherwise specified, all reproducible media, whether negative or positive, including original illustrative work, shall be deliverable items under the contract, and shall be delivered as specified in the contract; this does not include press plates (see 6.2).

3.4.7 Toxicity. - When used for intended purposes, the lubrication instruction media shall have no adverse affect on the health of using personnel. Any question regarding toxicity should be referred to the cognizant departmental medical authority.

3.4.8 Workmanship. - Lubrication instruction media furnished hereunder shall be manufactured in accordance with high grade commercial practice covering the class of work. The media shall be clean, well finished, and free from dirt, oil, foreign matter, or any defect which may affect their appearance or serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility. - Inspections are the responsibility of the requiring activity.

4.2 Proofs. - The contractor shall submit a single complete set of comprehensive proofs, as required by the contract (see 6.2) for the inspection and approval of the requiring agency. Submittal by increments is not permitted.

4.3 Material. - Sampling, inspection, and test of the material used in the construction of lubrication instruction media shall be in accordance with the requirements of applicable documents cited (see 2.1).

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. - Packaging and packing of lubrication instruction media shall be in containers complying with Uniform Freight Classification Rules, or with National Motor Freight Classification Rules, or with other carrier rules applicable to the mode of transportation. In addition, classified media shall be packaged and packed in accordance with the Industrial Security Manual for Safeguarding Classified Information (Attachment to DD Form 441).

5.2 Marking. - All shipping containers shall be marked in accordance with Standard MIL-STD-129, except that classified media shall be marked in accordance with the Industrial Security Manual for Safeguarding Classified Information (Attachment to DD Form 441).

6. NOTES

6.1 Intended use. - The requirements of this specification are intended to provide specific and concise method and data for use in the final production of lubrication instruction media.

6.2 Ordering data. - Procurement documents should furnish the following data:

a. Title, number, and date of this specification.

b. Type of lubrication media required (see 1.2).

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c. If Type II or III, specify if a graphic chart is not required (see 3.1.2 and 3.1.3).

d. If Type II or III, specify variations in content (see 3.1.2.1 and 3.1.3.1).

e. If Type I, III, IV, or V specify color of ink if other than black (see 3.3.5).

f. If Type II specify if white lines on dark background are required (see 3.4.2).

g. If Type II mounting is to be by screws or rivets, or adhesive, specify (see 3.4.2).

h. If Type III specify interior or exterior (see 3.4.3).

i. If Type III specify background color (see 3.4.3).

j. Shipping instructions for reproducible media (see 3.4.6).

k. The numerical designator (see 3.2.1.2).

l. The wording for title (see 3.2.1.3 and 3.2.5.1).

m. Shipping information for proofs (see 4.2).

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing Activity
Navy - Marine Corps

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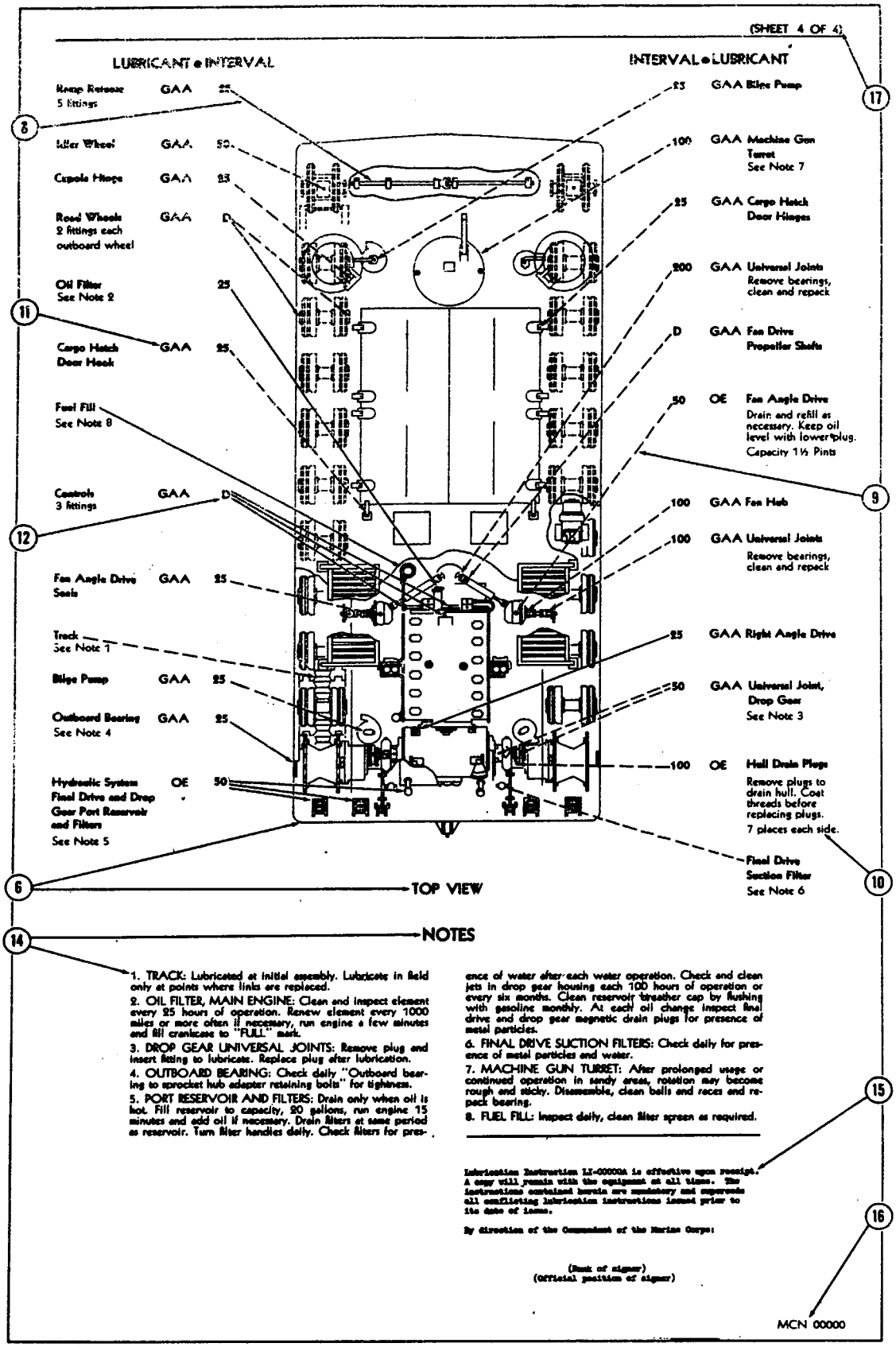


Figure 2. Last Page of Lubrication Chart

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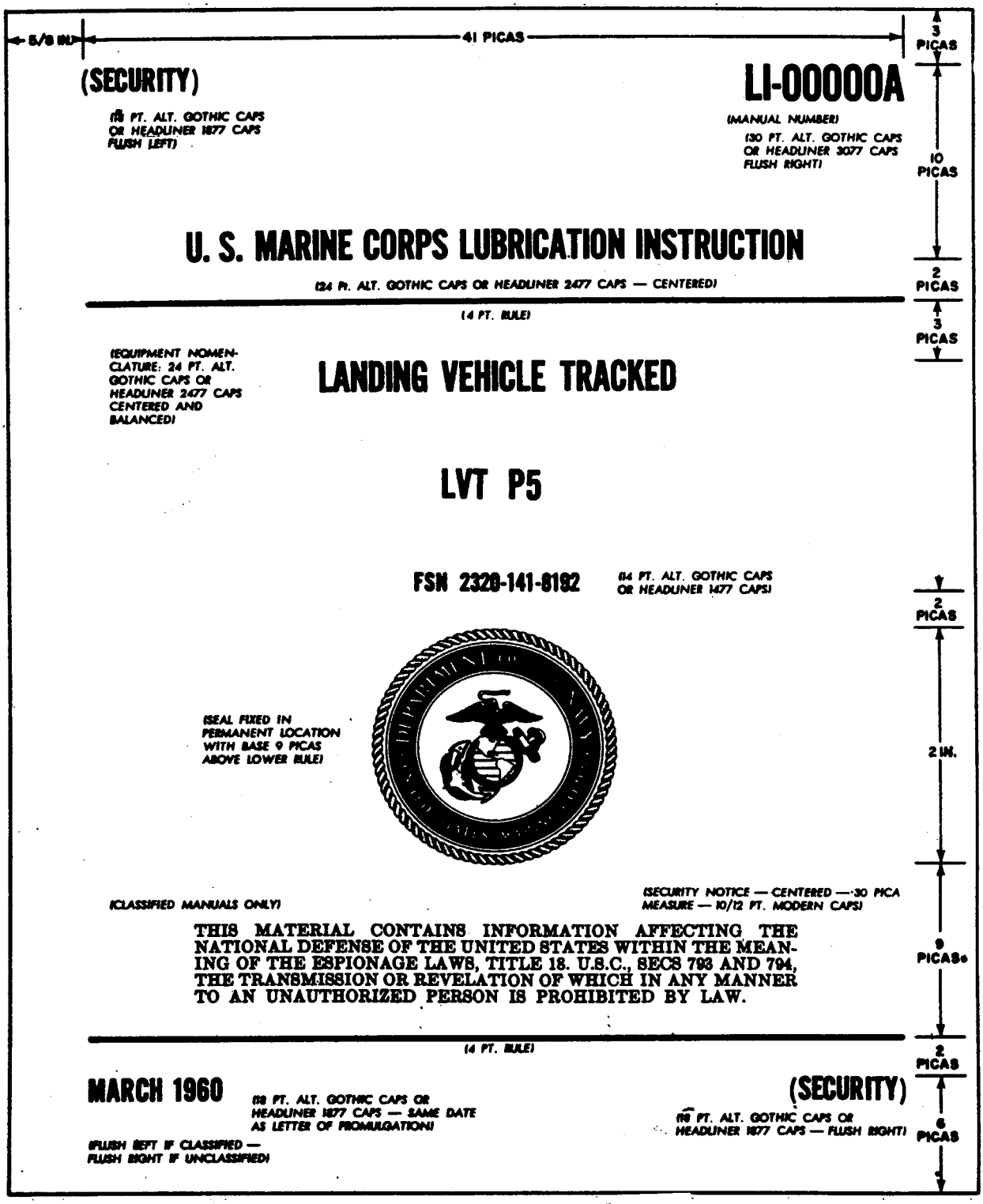
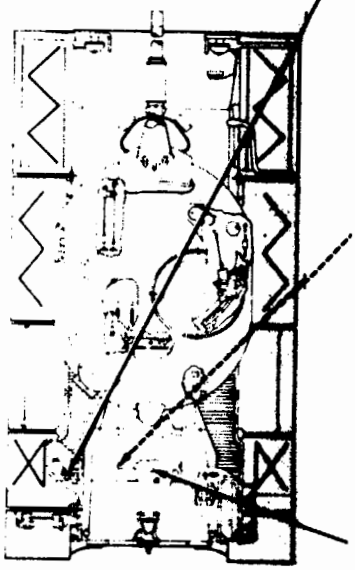


Figure 3 Front Cover

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LOCATION IN VEHICLE	LUBRICATION INSTRUCTIONS	LUBRICATION POINT



LUBRICANT • INTERVAL

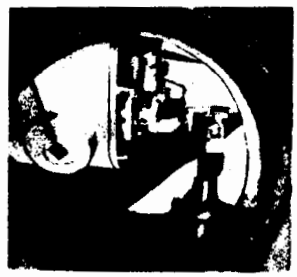
TRANSMISSION FILL AND LEVEL — OE D

(Check Level.)
(To check, open left rear grille door.)
(See Note.)



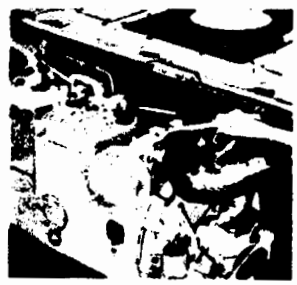
TRANSMISSION DRAIN — S

(Drain and Refill.)
(To drain, remove cover plate on bottom of hull.)
(See Note.)



TRANSMISSION OIL FILTER — Q

(Remove element, clean, dry, inspect, and install.)



NOTE

Check transmission oil level daily. If oil level is not up to FULL mark on gage, add oil. Set parking brake and start engine. Place selector lever in PARK position and idle 2 to 3 minutes to fill oil passages. With engine idling, transmission oil level should read at FULL AT ENGINE IDLE mark on gage. If not, add oil with engine idling, to obtain correct oil level. Drain every 750 miles or semi-annually. Drain only when hot after operation.

LUBRICATE DOTTED ARROW POINTS ON BOTH SIDES OF THE VEHICLE.

—KEY—

LUBRICANTS	EXPECTED TEMPERATURE			FOR ARCTIC OPERATION (refer to TM 9-207)	INTERVALS
	Above +32°F	+40 to −10°F	0°F to −65°F		
OE —LUBRICATING OIL, INTERNAL COMBUSTION ENGINE	OE 10	OE 10	OES		D – DAILY Q – QUARTERLY S – SEMI-ANNUALLY
OES—LUBRICATING OIL, INTERNAL COMBUSTION ENGINE, SUB-ZERO					

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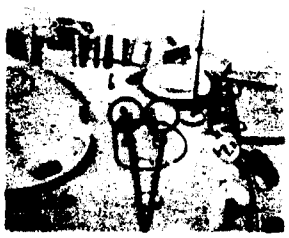
Figure 4. Lubrication Points Serviced From Engine Compartment.

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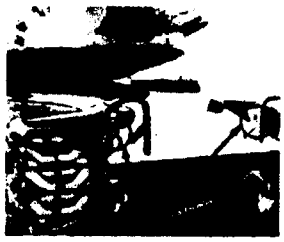
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OIL CAN POINTS

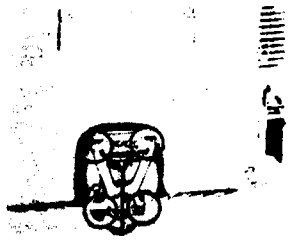
QUARTERLY, LUBRICATE WITH PL-LUBRICATING OIL, GENERAL PURPOSE
PL (MEDIUM) ABOVE +32 F PL (SPECIAL) BELOW +32 F



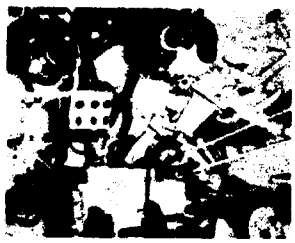
LOADER'S HATCH DOOR HINGES



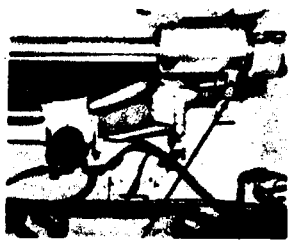
HEADLAMP REMOVAL NUTS



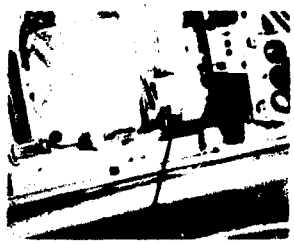
GUN TRAVELING LOCK



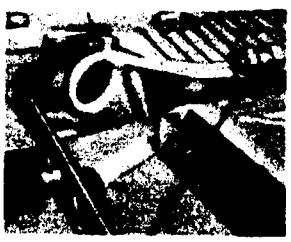
TRAVERSING HANDLE



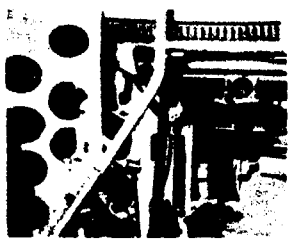
DRIVER'S HATCH LOCKS



BRAKE MECHANISM LINKAGE



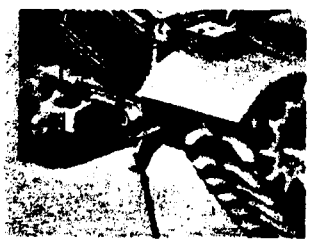
GRILLE DOOR HINGES AND RETAINING BOLTS



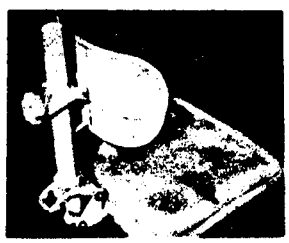
AMMUNITION RACK RETAINERS



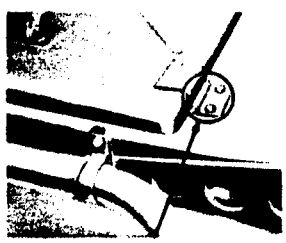
TRANSMISSION SUPPORT GUIDE RAILS AND ROLLERS



TOWING HOOKS



SEAT MOVING PARTS



FENDER STOWAGE BOX LATCHES AND HINGES

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Figure 5 Oil Can Points - Localized Views

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LUBRICATED AT TIME OF ASSEMBLY BY FIELD OR DEPOT MAINTENANCE



AZIMUTH INDICATOR



BRAKE LINKAGE



STARTER



SHIFTING LINKAGE



GENERATOR



TRANSMISSION MOUNTS

DO NOT LUBRICATE



AIR CLEANER BLOWER MOTOR



TRAVERSE LOCK



STARTER SOLENOID

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Figure 6 Materiel Lubricated by Field and Depot, and Materiel Not to be Lubricated - Localized Views

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